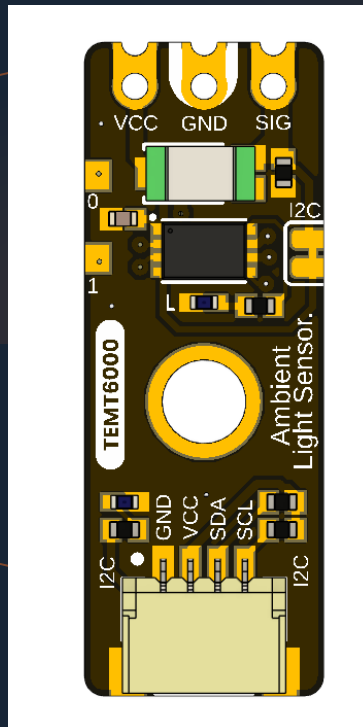


UNIT ELECTRONICS · DEVLAB ECOSYSTEM · ATOM FAMILY

UNIT ATOM TEMT6000

Version 1.1.0



TEMT6000 ambient-light module with an onboard I2C interface, Qwiic expansion, and direct analog access.

TEMT6000

I2C + ANALOG

QWIIC

DEVLAB ECOSYSTEM

ATOM FAMILY

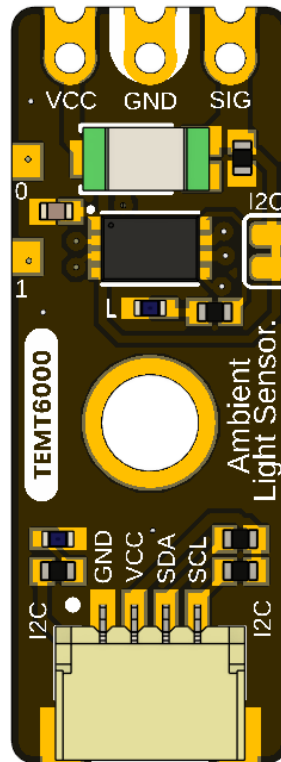
Contents

- **Description**
 - **Applications**
 - **DevLab Format Compatibility**
 - **Hardware Features**
- **1 The Board**
 - **1.1 Accessories**
 - **1.2 Board Identification**
 - **1.3 Identified Assemblies**
 - **1.4 Board Views**
 - **1.5 Handling**
- **2 Ratings**
 - **2.1 Current Module Operating Conditions**
 - **2.2 Interface Controller Characteristics**
 - **2.3 TEMT6000 Maximum Ratings**
 - **2.4 TEMT6000 Characteristics**
 - **2.5 Legacy Analog Circuit Scope**
 - **2.6 Unspecified Current-Module Characteristics**
 - **2.7 Electrical Precautions**
- **3 Functional Overview**
 - **3.1 Functional Block Diagram**
 - **3.2 Controller Clock and Execution Model**
 - **3.3 Measurement and Command Flow**
 - **3.4 Logical Register Association Map**
 - **3.5 I2C and DDP Register Model**
 - **3.6 Global Command Address Map**
 - **3.7 Command Register Descriptions**
 - **3.8 Response-Byte Interpretation**
 - **3.9 Direct Analog Mode**
 - **3.10 I2C Disable Bridge**
 - **3.11 Optical Response**
 - **3.12 Service and Expansion Signals**
- **4 Connectors & Pinouts**
 - **4.1 General Pinout**
 - **4.2 Direct Sensor Contacts**

- **4.3 Qwiic I2C Positions A and B**
- **4.4 Auxiliary and Factory Service Functions**
- **4.5 Indicators and I2C Bridge**
- **5 Board Operation**
 - **5.1 I2C Bring-up**
 - **5.2 Application 1: Read Ambient-Light Data**
 - **5.3 Application 2: Change the I2C Address**
 - **5.4 Application 3: Averaged Capture**
 - **5.5 Application 4: Built-in Indicator**
 - **5.6 Application 5: Direct Analog Bring-up**
 - **5.7 Service Operation: Modifying the I2C Bridge**
 - **5.8 Troubleshooting Guide**
- **6 Mechanical Information**
 - **6.1 Current Board Envelope**
 - **6.2 Integration Clearances**
 - **6.3 Mechanical Data Not Specified**
- **7 Company Information**
- **8 Reference Documentation**
- **9 Appendix**
 - **9.1 Legacy V0.0.1 Schematic**
 - **9.2 Document Control**
 - **9.3 Required Technical Releases**
 - **9.4 Source Notes and Inconsistencies**
 - **9.5 Current Firmware Limitations**

Description

The UNIT ATOM TEMT6000 is an I2C-compatible ambient-light sensor module from UNIT Electronics, the company that creates and develops the DevLab board ecosystem. Atom is a product family within that ecosystem. Hardware V0.3.1 combines a TEMT6000 visible-light phototransistor with an onboard interface controller, Qwiic I2C routing, and direct access to the sensor's analog signal. The fitted sensor manufacturer and exact orderable suffix have not been confirmed.



The released pinout identifies two optional Qwiic connection positions, direct VCC/GND/SIG contacts, reserved PA0 and PA1 pads, factory-only reset and SWD functions multiplexed on the I2C port, power and built-in indicators, and a solder bridge that disables the I2C function when cut. Inside the controller, the sensor ADC input is PA2, BUILTIN is driven by PB5, and the I2C bus uses PB6/SCL/SWCLK and PA10/SDA/SWDIO.

Applications

- I2C-connected ambient-light acquisition
- Direct analog light measurement during development and calibration
- Automatic display and indicator brightness control
- Day/night and relative-light detection
- Environmental data logging
- Educational I2C, ADC, and phototransistor experiments

DevLab Format Compatibility

As part of the DevLab Atom family, the TEMT6000 module uses a compact, standardized format for rapid prototyping and integration into sensing systems. It provides direct access to the analog signal through the VCC, GND, and SIG contacts, while the onboard controller publishes digital sensor readings through I²C. The module is intended for nominal 3.3 V or 5 V operation. The controller guidance uses a conservative 2.0–5.5 V range; complete-module electrical limits require the current schematic and validation.

The board has two optional positions for horizontal 4-pin, 1.00 mm-pitch JST/Qwiic connectors carrying GND, VCC, SDA, and SCL. They provide alternative I²C connection points. Before connection, confirm the populated connector and its orientation, and ensure the host tolerates the actual I²C pull-up voltage; a 3.3 V host may require level translation when the module is powered at 5 V. The pull-up resistance is not specified by the released documentation.

Hardware Features

- TEMT6000 ambient-light phototransistor; fitted manufacturer/orderable part not confirmed
- 32-bit Arm Cortex-M0+ controller with 16 KB Flash and 2 KB SRAM, using its internal HSI at up to 24 MHz; no external oscillator is used
- Nominal 3.3 V and 5 V operation; controller upper operating limit 5.5 V
- 7-bit I2C slave operation from 100 kHz through 400 kHz
- Qwiic GND, VCC, SDA, and SCL routing
- Direct analog SIG access with adjacent VCC and GND
- Reserved PA0 and PA1 pads with no current application assignment
- Internal sensor ADC connection to controller pin PA2
- Shared PB6 / SCL / SWCLK / PA10 / SDA / SWDIO controller mapping
- Factory-only SWD/reset functions, power LED, and PB5-controlled built-in LED
- Cuttable solder bridge for disabling I2C operation
- Manufacturer Part Number (MPN) UE0098

The controller implements DevLab Device Protocol (DDP) v1.0. The TEMT6000 profile is identified by Device ID 0x0102; command TEMT6000_RAW (0x80) returns a 12-bit ADC sample in an unsigned 16-bit little-endian response. The current controller firmware reports factory address 0x20, firmware/hardware 1.0, and capabilities 0x000001B9. Procurement does not establish the fitted controller suffix, so the published controller voltage guidance uses the conservative x7 range without identifying the physical variant. The technical package does not provide the exact controller package or ordering suffix, current-revision schematic, or complete module electrical limits.

SWDIO shares physical pin PA10 with SDA, and SWCLK shares PB6 with SCL; there is no separate SWD port and the two modes are mutually exclusive. SWD/reset are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.

1 The Board

The design supports two host paths. A Qwiic-capable controller can attach through the I2C positions, while an ADC-capable host or test instrument can sample the exposed SIG contact directly. Factory debug reuses the same controller pins and physical port as I2C; it is not a separate user interface.

1.1 Accessories

No accessory bundle is specified. Typical integration items are:

Accessory	Purpose	Selection notes
Qwiic cable	Connects GND, VCC, SDA, and SCL	Verify 1.0 mm pitch, orientation, and contact order; connectors are shown as optional
I2C-capable host	Scans and communicates with the module	Use 7-bit addressing and a clock from 100 kHz through 400 kHz
Analog test lead or carrier	Accesses VCC, GND, and SIG	SIG must connect to a voltage-compatible ADC input
Logic analyzer	Checks I2C activity	Use input thresholds compatible with the powered board
Reference lux meter	Supports optical calibration	Required for quantitative illuminance validation

1.2 Board Identification

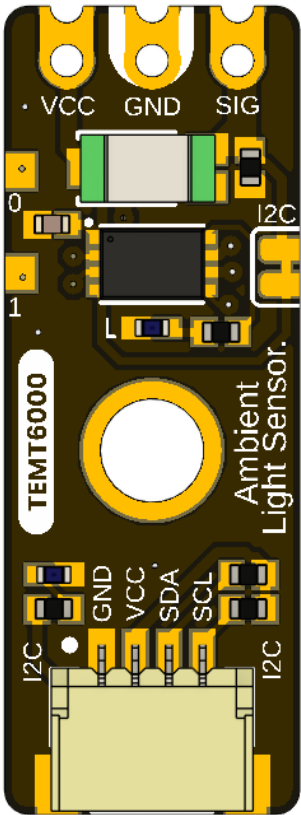
Item	Value
Product	UNIT ATOM TEMT6000 Ambient Light Sensor
Brand / company	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Product type	I2C-compatible and direct-analog ambient-light module
Optical component	TEMT6000; fitted manufacturer/orderable part not confirmed
Interface controller	32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM; fitted suffix not asserted
Manufacturer Part Number (MPN)	UE0098
Current board artwork	
Available schematic	Legacy analog hardware V0.0.1 only
Product Reference	Version 1.1.0

Board, pinout, schematic, and documentation revisions are controlled independently.

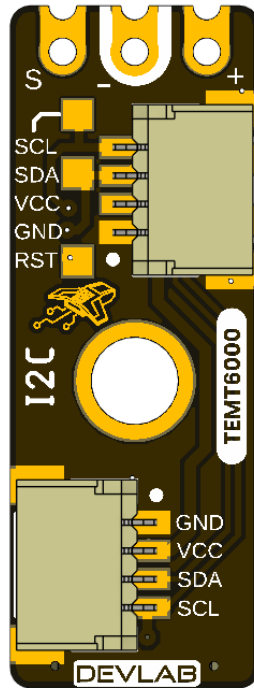
1.3 Identified Assemblies

Assembly	Function	Source status
TEMT6000 sensor	Converts visible light to photocurrent	Functional identity shown by board/pinout artwork; manufacturer and exact suffix unconfirmed
Interface controller	Samples or processes the sensor for I2C access	32-bit Arm Cortex-M0+; 16 KB Flash and 2 KB SRAM; fitted suffix not asserted; published voltage guidance uses the x7 range
Qwiic positions A and B	I2C power and bus access	Shown as optional horizontal JST connectors
Direct contacts	VCC, GND, and analog SIG	Identified on the top view
I2C disable bridge	Disconnects or disables I2C when cut	Function identified; exact circuit unspecified
Power and built-in LEDs	Power and firmware indication	BUILT IN is driven by controller PB5
PA0/PA1 and service functions	Reserved GPIO and factory reset/debug	PA0/PA1 have no current application; SWD aliases share the I2C port
Internal controller mapping	PA2 ADC, PB5 built-in actuator, PB6/SCL / SWCLK, PA10/SDA/SWDIO	Current firmware/hardware mapping

1.4 Board Views



The top view shows the direct sensor contacts, controller, sensor, mounting hole, indicator circuitry, and one Qwiic connector position.



The bottom view labels Qwiic, the SWD aliases on that same port, reset, and the second optional connector position.

1.5 Handling

Use normal ESD precautions. Keep the transparent sensor package clean and optically unobstructed. Remove power before changing connectors or modifying the I2C solder bridge. Do not attach an SWD probe or attempt to replace the firmware. SWD/reset access is reserved for the manufacturer and advanced factory diagnostics; I2C must be inactive and isolated before factory SWD use.

2 Ratings

No current-revision schematic or released board-level electrical table is present for V0.3.1. This chapter therefore separates the expected Qwiic integration domain from comparative TEMT6000X01 reference data and explicitly marks current module values that still require release. That external data does not identify the manufacturer or exact part fitted to the board.

2.1 Current Module Operating Conditions

Parameter	Value	Status
Nominal module supply	3.3 V or 5 V	Supported operating points; use a current-limited source during bring-up
Published controller operating voltage	2.0–5.5 V	Conservative x7 voltage range; supports nominal 3.3 V and 5 V without asserting the fitted suffix
I2C logic domain	Follows the actual bus pull-up rail	At 5 V, use a 5 V-tolerant host or bidirectional level translation
I2C bus speed	100 kHz to 400 kHz	Supported operating range
I2C addressing	7-bit slave	Valid configurable addresses are 0x08 . . 0x77
Factory I2C address	0x20	Pass the 7-bit value, not wire byte 0x40
Device protocol	DDP v1.0	Command transaction followed by an exact-length read transaction
Logical Device ID	0x0102	TEMT6000 identity; independent of I2C address
Firmware / hardware	1.0 / 1.0	Current observable controller profile
Capability bitmap	0x000001B9	I2C config, analog input, sensor data, relay, watchdog, persistent config
Raw digital sample	0 to 4095	12-bit ADC code returned as unsigned 16-bit little endian
ADC update interval	Approximately 20 ms	Background acquisition; I2C read returns latest published sample
Command processing delay	2 to 5 ms	Use 5 ms conservatively before reading
Pending setter timeout	250 ms	Parameter must arrive before expiry
Direct SIG range	Not specified	Requires current analog-stage schematic and measurement
Module current consumption	Not specified	Controller, LEDs, pull-ups, and sensor load are undocumented
Module ambient temperature	Not specified	No complete-module qualification supplied

The controller's 2.0–5.5 V rating confirms that the controller itself can operate at 5 V; it does not by itself validate 5 V operation for every circuit on the module. The direct SIG path and I2C levels are referenced to the powered system, so every attached host must tolerate the resulting voltage or use level translation. Complete-module absolute maximums, pull-up resistance, and current consumption still require the schematic and qualification.

2.2 Interface Controller Characteristics

The supplied UE0102 reference manual identifies a PY32F003L24D6TR and publishes 1.7–5.5 V and –40 to +85 °C for that separate development board. It does not establish which ordering suffix was procured for this UE0098 assembly. This Product Reference therefore does not identify the fitted ordering suffix; it uses the x7 range of 2.0–5.5 V as conservative voltage guidance.

Feature	Controller capability
CPU and application clock	32-bit Arm Cortex-M0+; internal HSI at up to 24 MHz; external HSE oscillator not used
Memory	16 KB Flash and 2 KB SRAM
Published operating voltage	2.0–5.5 V, conservative x7 range
ADC	One 12-bit ADC, up to 10 external channels, conversion range 0 . . VCC
I2C	Standard mode 100 kHz, Fast mode 400 kHz, 7-bit addressing
GPIO	Up to 18 I/Os, all available as external interrupts
Timers	TIM1, TIM3, TIM14, TIM16, TIM17, LPTIM, IWDG, WWDG, SysTick, IRTIM
Other interfaces	SPI, two USARTs, DMA, RTC, CRC-32, two comparators, UID; SWD is reserved for factory use on this module

The controller range applies to the controller only. It does not establish the pull-up resistance, complete-module current, or absolute maximum of every external contact.

2.3 TEMT6000 Maximum Ratings

The values below are included only as a comparative TEMT6000 profile. They are not proof that a Vishay part is fitted, are not guaranteed for the fitted sensor, and do not define limits for the controller, LEDs, pull-ups, connectors, or complete board.

Parameter	Symbol	Value	Unit
Collector-emitter voltage	VCE0	6	V
Emitter-collector voltage	VECO	1.5	V
Collector current	IC	20	mA
Power dissipation at 25 °C	PV	100	mW
Junction temperature	Tj	100	°C
Component operating temperature	Tamb	–40 to +100	°C

2.4 TEMT6000 Characteristics

Unless noted otherwise, the values are specified at 25 °C and are provided for comparison only. Production specifications must come from the confirmed fitted part and module-level validation.

Parameter	Test condition	Min.	Typ.	Max.	Unit
Collector dark current	VCE = 5 V, Ev = 0	—	3	50	nA
Collector light current	Ev = 20 lx, CIE illuminant A, VCE = 5 V	3.5	10	16	μA
Collector light current	Ev = 100 lx, CIE illuminant A, VCE = 5 V	—	50	—	μA
Collector-emitter capacitance	VCE = 0 V, f = 1 MHz, dark	—	16	—	pF
Collector-emitter saturation voltage	Ev = 20 lx, IPCE = 1.2 μA	—	0.1	—	V
Angle of half sensitivity	—	—	±60	—	degrees
Peak sensitivity wavelength	—	—	570	—	nm
Spectral bandwidth at half sensitivity	—	440	—	800	nm

2.5 Legacy Analog Circuit Scope

The V0.0.1 schematic shows a TEMT6000 with a 10 kΩ emitter resistor, giving the first-order relation $V_{\text{SIGNAL}} \approx IPCE \times 10 \text{ k}\Omega$ outside saturation. Applying the 100 lx typical current from the datasheet predicts about 0.50 V; this is not a guaranteed current-board value.

V0.3.1 adds an interface controller and other circuitry. Without its schematic, the legacy resistor value and transfer function must not be represented as a guaranteed current-board characteristic. Direct SIG must be measured and validated on V0.3.1.

2.6 Unspecified Current-Module Characteristics

- Complete-module absolute maximum, current consumption, and power-up behavior
- Exact oscillator/timing tolerance and electrical bus-loading limits
- Pull-up resistance, bus capacitance allowance, and level compatibility
- Direct analog transfer function, load resistance, range, accuracy, and source impedance
- Guaranteed lux range, accuracy, repeatability, response time, and calibration
- Exact controller package/ordering suffix, released firmware image, and update procedure
- Board-level temperature, ESD, EMC, humidity, and ingress ratings

2.7 Electrical Precautions

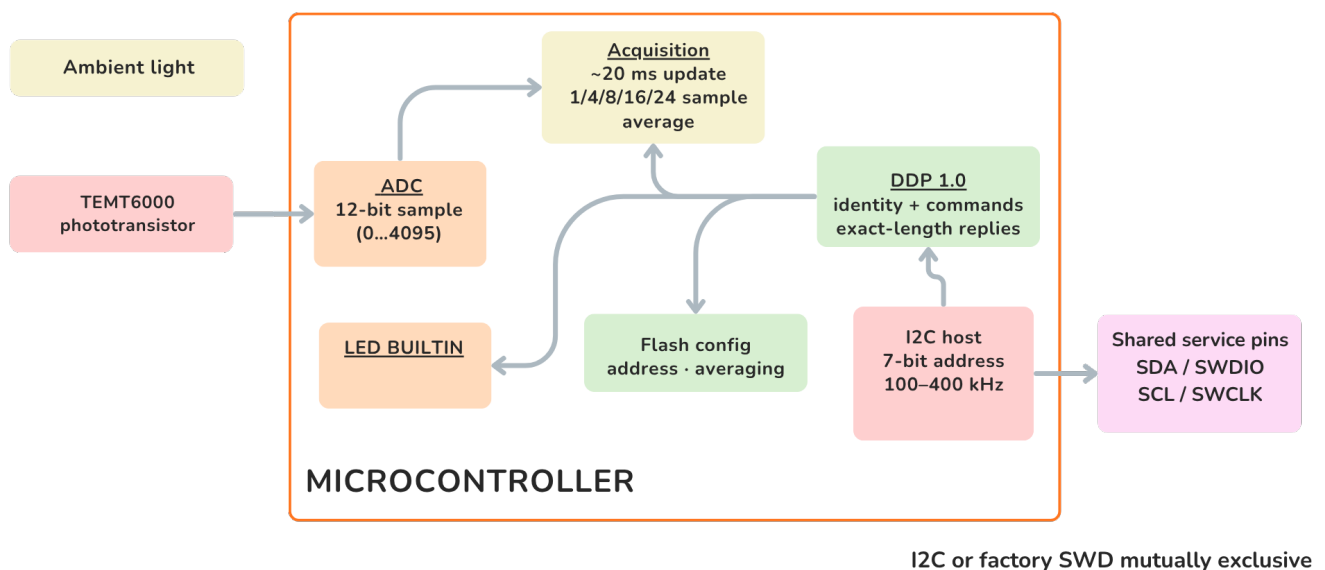
1. Use a current-limited 3.3 V or 5 V supply for engineering bring-up.
2. Verify connector orientation and share ground before applying power.
3. Do not exceed the controller upper operating limit of 5.5 V.
4. At 5 V, confirm host tolerance at SDA, SCL, and SIG, or add suitable level translation before connection.
5. Do not connect to reset or SWD; these signals are reserved for manufacturer programming and advanced factory diagnostics, not user firmware replacement.
6. Remove power before cutting or reworking the I2C solder bridge.
7. Confirm the current schematic and interface specification before production.

3 Functional Overview

The TEMT6000 sensor generates a light-dependent analog signal. That signal is exposed at SIG and acquired by the onboard controller through PA2/ADC0 for DDP access. The controller uses its internal 24 MHz clock; this application does not use an external oscillator. Controller output PB5 drives the BUILTIN LED.

The missing schematic must still confirm the passive sensor circuitry and complete module electrical characteristics. The logical associations and observable firmware behavior documented below are the released interface.

3.1 Functional Block Diagram



PA0 and PA1 are reserved with no current application assignment · RESET/SWD are factory-only.

Figure 3.1 — Functional signal and control paths. The direct SIG branch and controller PA2 ADC observe the sensor path. Firmware averages the sample and publishes it through DDP over I2C. The internal HSI supplies the 24 MHz application clock; no HSE crystal or external clock is used.

3.2 Controller Clock and Execution Model

Property	Application configuration	Functional consequence
Controller core	32-bit Arm Cortex-M0+	Executes acquisition, DDP, configuration, and indicator control
Controller memory	16 KB Flash and 2 KB SRAM	Memory configuration for this module
Maximum application clock	24 MHz	Maximum clock used and documented for this product firmware
System-clock source	Internal high-speed oscillator (HSI)	Self-clocked controller operation
External high-speed clock (HSE)	Not used	No external oscillator or crystal is required by this application
ADC execution	Background task, approximately every 20 ms	Reads return the most recently published sample rather than starting a conversion
Command execution	Command is prepared after an I2C write	Host waits 2–5 ms before the exact-length read

The 24 MHz value describes this application configuration. It must not be replaced by a higher generic family maximum when specifying this product.

3.3 Measurement and Command Flow

1. Ambient light changes the TEMT6000 analog output.
2. Controller input PA2/ADC0 performs a 12-bit conversion.
3. Firmware discards one settling conversion, updates the circular sample buffer, and publishes the rounded average.
4. A host addresses the module and writes one DDP command byte.
5. After STOP and a 2–5 ms processing interval, the host reads the exact response length.
6. READ_ADC0 and TEMT6000_RAW return the same latest published sample.

Functional timing	Value	Notes
ADC update interval	Approximately 20 ms	About 50 background updates per second
Averaging selections	1, 4, 8, 16, or 24 samples	Persistent configuration
Full averaging-window fill	$N \times 20 \text{ ms}$	20, 80, 160, 320, or 480 ms
I2C bus clock	100 kHz to 400 kHz	Standard mode and Fast mode
Command-to-read delay	2–5 ms	Use 5 ms conservatively
Staged-parameter timeout	250 ms	Applies to 0x21, 0x62, and 0xA3

3.4 Logical Register Association Map

DDP presents command-selected data fields rather than MCU memory addresses. The following table associates each externally visible logical field with its physical resource and command address.

Logical field	Physical association	DDP command address	Data or behavior
Device identity	Firmware constant	0x00	Device ID 0x0102
Firmware version	Firmware metadata	0x01	Version 1.0
Hardware version	Product profile	0x02	Version 1.0
Capabilities	Firmware feature bitmap	0x03	0x000001B9
Protocol version	DDP implementation	0x04	DDP 1.0
Active I2C address	I2C slave configuration	0x20	Factory 0x20; configurable 0x08 . . 0x77
Persistent configuration	Internal Flash	0x21 . . 0x24, 0x62, 0xA3	Address, averaging, and toggle time
ADC0	TEMT6000 signal on controller PA2	0x60	Unsigned 12-bit ADC code
ADC averaging	Circular sample buffer	0x62, 0x63	1, 4, 8, 16, or 24 samples
TEMT6000 sensor data	Same published PA2/ADC0 sample	0x80	Same two-byte response as 0x60
Built-in indicator	Controller PB5/BUILTIN	0xA0 . . 0xA4	Off, on, toggle, and persistent pulse time
System reset	Controller reset path	0xF0	Resets without a response read
I2C service pins	PA10/SDA, PB6/SCL	Transport, not a data register	Shared with factory-only SWD aliases
Reserved contacts	PA0, PA1	None	No current application assignment

3.5 I2C and DDP Register Model

DDP is **not** a memory-mapped register interface. The first byte written after the 7-bit slave address is a command selector analogous to a register address. The command byte selects a prepared response or an operation; it does not expose the PY32F003 memory map.

The bus address and command address are independent values. A host passes the **7-bit I2C slave address** (0x20 at the factory) to its library, then writes one **DDP command address** from 0x00 through 0xFF. Do not pass shifted wire bytes 0x40/0x41 as the device address.

```
host → [7-bit slave address + W] [DDP command address] → STOP
      wait 2...5 ms
host ← [7-bit slave address + R] [exact response length] ← STOP
```

Multi-byte values are little endian. Staged setters require a second one-byte write transaction within 250 ms; the selector and parameter must not be sent in the same I2C write transaction.

3.6 Global Command Address Map

DDP address block	Commands defined for this profile	Current support
0x00 . . 0x1F Device information	0x00 . . 0x04	Identity and discovery registers implemented
0x20 . . 0x3F Configuration	0x20 . . 0x24	I2C address and persistent configuration implemented
0x40 . . 0x5F Digital I/O	None	Not implemented on this module
0x60 . . 0x7F Analog input	0x60, 0x62, 0x63	ADC0/PA2 and averaging implemented; 0x61 unsupported
0x80 . . 0x9F Sensor data	0x80	TEMT6000 raw sample implemented
0xA0 . . 0xBF Actuators	0xA0 . . 0xA4	PB5/BUILTIN indicator control implemented
0xC0 . . 0xDF Calibration	None	Capability not announced; unsupported
0xE0 . . 0xEF Reserved	None	Reserved; do not use
0xF0 . . 0xFF System	0xF0 . . 0xF4	Reset implemented; watchdog experimental; remaining functions incomplete or unsupported

Any undefined or unsupported selector returns the current packed unknown-command response with low nibble 0xB.

3.7 Command Register Descriptions

Access notation: R selects and reads a prepared response; W performs an operation or begins a staged parameter write; W/R writes the selector and then reads its acknowledgement or result.

3.7.1 Device Information Registers

Address	Symbol	Access	Exact response	Association
0x00	GET_DEVICE_ID	R	[0x02, 0x01]	Device ID 0x0102
0x01	GET_FIRMWARE_VERSION	R	[0x01, 0x00]	Firmware major 1, minor 0
0x02	GET_HARDWARE_VERSION	R	[0x01, 0x00]	Hardware major 1, minor 0
0x03	GET_CAPABILITIES	R	[0xB9, 0x01, 0x00, 0x00]	Little-endian bitmap 0x000001B9
0x04	GET_PROTOCOL	R	[0x01, 0x00]	DDP major 1, minor 0

Discovery order is 0x04, 0x00, 0x01, 0x02, then 0x03. A host must verify protocol major 1 and Device ID 0x0102; the I2C address alone does not identify the product.

3.7.2 Configuration Registers

Address	Symbol	Access	Parameter or response	Association
0x20	GET_I2C_ADDR	R	One-byte active 7-bit address	I2C peripheral address
0x21	SET_I2C_ADDR	W	Staged 0x08 . . 0x77; ACK low nibble 0xD	Saves Flash and resets after final ACK
0x22	SAVE_CONFIG	W/R	Returns 0x00	Idempotent; setters already persist
0x23	RESET_FACTORY	W/R	ACK low nibble 0xE	Restores defaults; separate reset required
0x24	GET_I2C_STATUS	R	0x00 default; 0x01 loaded from Flash	Address-source status

3.7.3 Input and Sensor Registers

Address	Symbol	Access	Parameter or response	Physical association
0x60	READ_ADC0	R	[LSB, MSB], valid code 0 . . 4095	Latest PA2 sample
0x62	SET_ADC_AVERAGING	W	Staged 1, 4, 8, 16, or 24; ACK low 0xC	Persistent averaging depth
0x63	GET_ADC_AVERAGING	R	One byte: 1, 4, 8, 16, or 24	Active averaging depth
0x80	TEMT6000_RAW	R	[LSB, MSB], valid code 0 . . 4095	Same published sample as 0x60

READ_GPIO0/1 (0x40/0x41), WRITE_GPIO0/1 (0x42/0x43), and READ_ADC1 (0x61) are unsupported.

3.7.4 Indicator and System Registers

Address	Symbol	Access	Parameter, response, or effect	Physical association
0xA0	RELAY_OFF	W/R	ACK low nibble 0x0; output LOW	PB5/BUILTIN off
0xA1	RELAY_ON	W/R	ACK low nibble 0x1; output HIGH	PB5/BUILTIN on
0xA2	RELAY_TOGGLE	W/R	ACK low nibble 0x6; non-blocking pulse	PB5/BUILTIN toggle
0xA3	SET_TOGGLE_TIME	W	Staged 1 . . 40; ACK low nibble 0x7	Persistent units of 25 ms
0xA4	GET_TOGGLE_TIME	R	One byte 1 . . 40	Active pulse time
0xF0	RESET	W	Immediate reset; do not request a reply	Controller reset path

Address	Symbol	Access	Parameter, response, or effect	Physical association
0xF1	WATCHDOG_RESET	W	Experimental; no active hardware IWDG	Firmware system block
0xF2 . . 0xF4	Reset-info/NRST functions	—	Unsupported or incomplete	Reserved for future profile completion

3.8 Response-Byte Interpretation

Response bytes have meaning only within the selected command and expected length. For example, 0x01 may mean firmware major 1, hardware major 1, protocol major 1, the high byte of Device ID 0x0102, or “address loaded from Flash” from GET_I2C_STATUS. It is not a universal success code.

3.9 Direct Analog Mode

The top-side SIG contact provides direct sensor access for ADC experiments, production test, or calibration. Its transfer function and safe input range are not documented. Configure the host as a high-impedance analog input and measure the actual range before connecting a lower-voltage ADC.

The controller samples this sensor path internally on PA2, mapped to logical ADC0. READ_ADC0 and TEMT6000_RAW are equivalent in the current firmware.

3.10 I2C Disable Bridge

The released pinout instructs the user to cut a solder bridge to disable I2C. The missing current schematic leaves the exact isolation boundary unspecified. Do not assume that cutting the bridge removes pull-ups, disconnects every controller pin, or changes the analog transfer in a particular way. Rework only with power removed and after continuity checks on the target revision.

3.11 Optical Response

Reference-only TEMT6000X01 data lists a 570 nm peak, a 440 nm to 800 nm half-sensitivity spectral bandwidth, and a $\pm 60^\circ$ half-sensitivity angle. These values are comparative, not guaranteed for the unconfirmed fitted part. Source spectrum, temperature, sensor spread, enclosure windows, and mechanical shadowing can also change the response.

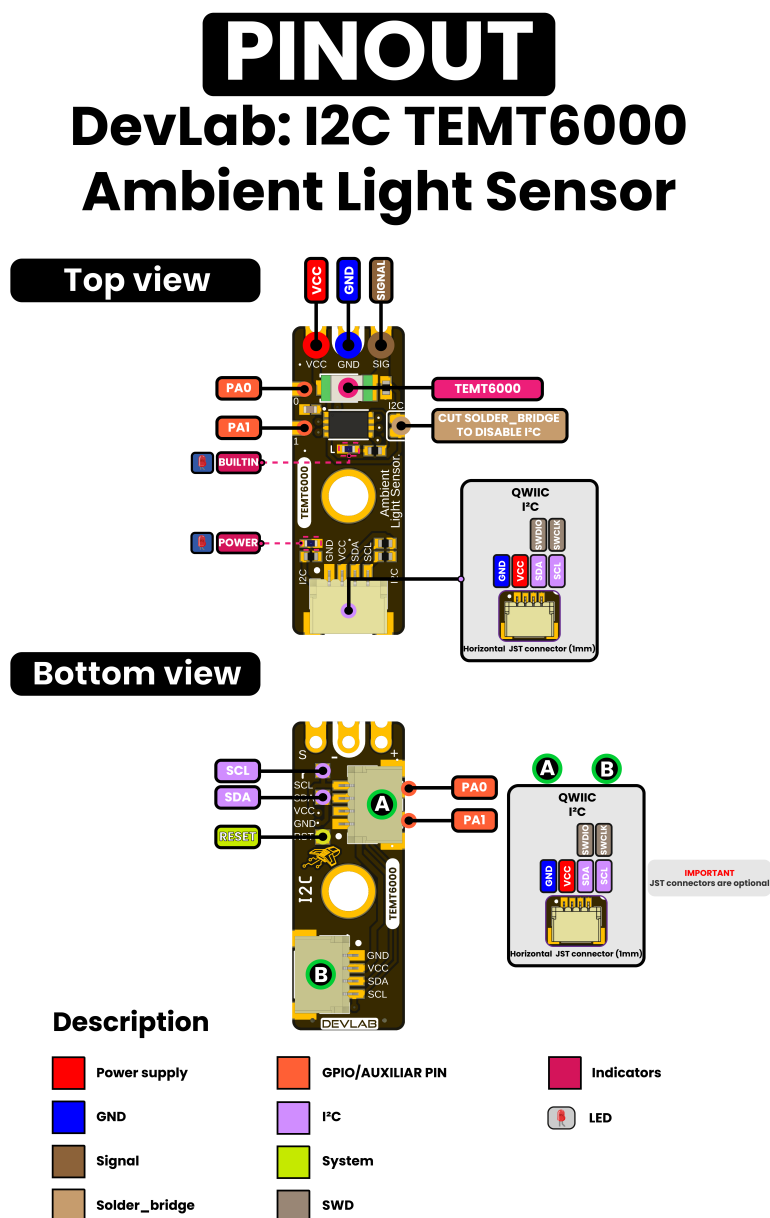
3.12 Service and Expansion Signals

The pinout exposes PA0, PA1, RESET, SWDIO, and SWCLK. PA0 and PA1 have no application assignment and are reserved for future use. SWDIO and SWCLK are alternate functions of the same PA10/PB6 pins used for SDA/SCL, not a separate port. They and RESET are reserved for manufacturer programming and advanced factory diagnostics; user firmware replacement is unsupported.

4 Connectors & Pinouts

Pin labels below come from the released English pinout. It does not provide controlled contact numbers or a mating-connector part number, so the signal order must be checked against connector orientation before producing a harness.

4.1 General Pinout



4.2 Direct Sensor Contacts

Label	Type	Description
VCC	Power	Module supply; nominal 3.3 V or 5 V operation
GND	Power	Common power and signal reference

Label	Type	Description
SIG / SIGNAL	Analog	Direct light-dependent sensor signal; current transfer function unspecified

These three contacts are shown at the end opposite the primary Qwiic connector.

4.3 Qwiic I2C Positions A and B

Signal	Type	Description
GND	Power	Common return
VCC	Power	I2C peripheral supply; nominal 3.3 V or 5 V
SDA / SWDIO	Bidirectional I2C / factory debug data	One physical PA10 line; normal use is SDA; host must tolerate the bus pull-up voltage
SCL / SWCLK	I2C clock / factory debug clock	One physical PB6 line; normal use is SCL at 100 kHz to 400 kHz

The pinout shows two possible horizontal 1.0 mm JST connector positions and marks JST connectors as optional. Verify which position is populated on the ordered assembly. Parallel connectors are intended for bus pass-through, but that topology must be confirmed by the current schematic.

I2C and SWD are multiplexed on this same physical port: PA10 is SDA/SWDIO, and PB6 is SCL/SWCLK. They are mutually exclusive, and there is no independent SWD connector or pin pair.

4.4 Auxiliary and Factory Service Functions

Label	Documented role	Qualification status
PA0	GPIO / reserved pin	No current application assignment; electrical limits unspecified
PA1	GPIO / reserved pin	No current application assignment; electrical limits unspecified
RESET	System reset	Factory service only; not intended for routine user operation

Do not connect an SWD probe or drive reset/debug functions. They are not user interfaces, and the product is not designed for user firmware replacement. Manufacturer diagnostics must make I2C inactive and isolate other bus devices before selecting SWD on the shared pins.

4.5 Indicators and I2C Bridge

The pinout identifies a POWER LED, a firmware-controlled BUILTIN LED, and a cuttable bridge used to disable I2C. Controller pin PB5 drives BUILTIN. Indicator polarity/current and bridge net connections still require the current schematic. The sensor ADC input is mapped internally to controller PA2; it is not an additional external contact in this pinout.

5 Board Operation

The repository supports safe electrical bring-up through I2C scanning, DDP identity and measurement reads, and direct analog observation.

5.1 I2C Bring-up

1. Inspect the board revision, connector population, and I2C bridge.
2. With power off, select a nominal 3.3 V or 5 V supply and connect GND, VCC, SDA, and SCL in the documented orientation. For 5 V operation, verify host tolerance at the bus pull-up voltage or add level translation.
3. Power from a current-limited source and check for unexpected heating.
4. Select an I2C clock from 100 kHz through 400 kHz; maintained examples use 400 kHz. The ESP32 and MicroPython examples map host GPIO6 to SDA and GPIO7 to SCL. Start at factory 7-bit address 0x20, or scan if it was changed.
5. Record the active address, protocol, firmware, hardware, and capability values reported during discovery.
6. Verify protocol/firmware/hardware 1.0 and capabilities 0x000001B9, then read either READ_ADC0 (0x60) or equivalent TEMT6000_RAW (0x80).
7. Use a logic analyzer to capture start, address, acknowledge, STOP, and read timing if enumeration fails.

The maintained DDP clients are `software/examples/i2c/cpp_examples/ddp_temt6000/ddp_temt6000.ino` and `software/examples/i2c/micropython/ddp_temt6000.py`. The MicroPython example imports the reusable `software/examples/i2c/micropython/lib/devlab_ddp.py` module, which is installed as `/lib/devlab_ddp.py` on the host. Low-level scanner sketches are `software/examples/i2c/cpp_examples/i2c_scanner/i2c_scanner.ino` and `software/examples/i2c/micropython/i2c_scan.py`.

5.2 Application 1: Read Ambient-Light Data

1. Initialize I2C at 100 kHz or 400 kHz using 7-bit address 0x20.
2. Run the Chapter 3 discovery sequence and accept only DDP major 1 with Device ID 0x0102.
3. Write TEMT6000_RAW (0x80), issue STOP, wait 5 ms, and read exactly two bytes.
4. Decode `raw = response[0] | (response[1] << 8)` and reject values above 4095.
5. Repeat no faster than the approximately 20 ms background update interval when a new sample is required.

READ_ADC0 (0x60) may replace TEMT6000_RAW; both access the same published PA2/ADC0 sample. The value is an uncalibrated ADC code, not lux. Convert to lux only after calibration against a reference meter in the final optical and mechanical installation.

5.3 Application 2: Change the I2C Address

The common DDP block provides GET_I2C_ADDR (0x20), SET_I2C_ADDR (0x21), SAVE_CONFIG (0x22), RESET_FACTORY (0x23), and GET_I2C_STATUS (0x24). Factory address is 0x20; a new value must be within 0x08..0x77.

Address change is a staged operation. Send 0x21, wait/read ACK low nibble 0xD, send the new one-byte value within 250 ms, wait about 100 ms, and read the final ACK at the old address. The setter saves Flash and reset follows when the final ACK is consumed. Wait about 300 ms and identify 0x0102 at the new address.

SAVE_CONFIG is currently idempotent. Factory restore returns ACK low 0xE but requires a separate reset before address 0x20 becomes active.

5.4 Application 3: Averaged Capture

The controller updates ADC0 in the background about every 20 ms and publishes a rounded circular-buffer mean. GET_ADC_AVERAGING (0x63) returns 1, 4, 8, 16, or 24. SET_ADC_AVERAGING (0x62) is staged with ACK low nibble 0xC and persists to Flash. Read before writing, and wait $N \times 20$ ms for a full window. Recommended values are 8 for UI/general automation and 16 for stable ambient measurement. Do not poll faster than 20 ms expecting a new value.

5.5 Application 4: Built-in Indicator

Controller PB5/BUILTIN uses the relay-compatible commands: RELAY_OFF (0xA0), RELAY_ON (0xA1), and non-blocking RELAY_TOGGLE (0xA2). It is not a digital GPIO signal, and the digital-I/O commands are unsupported. Pulse time is configured with SET_TOGGLE_TIME (0xA3) in 1 . . 40 units of 25 ms.

5.6 Application 5: Direct Analog Bring-up

1. Leave the I2C/service pins undriven and configure a host ADC as a high-impedance input.
2. Connect the direct GND, VCC, and SIG contacts with power removed.
3. Verify the actual SIG range using a multimeter before relying on the ADC; the controller ADC conversion range is 0 . . VCC.
4. Compare covered and illuminated readings and check that the response is repeatable.
5. Calibrate against a reference meter for quantitative lux measurements.

The analog examples remain available under `software/examples/adc/`, separate from the I2C clients. Their voltage calculations require measurement because the only available schematic describes legacy V0.0.1 hardware.

5.7 Service Operation: Modifying the I2C Bridge

Do not cut the bridge as a first troubleshooting step. First verify power, connector order, pull-ups, bus ownership, and scan timing. If analog-only operation requires isolation, remove power, document the original bridge, cut only the marked feature, inspect for debris, and confirm continuity before repowering. Restoring the bridge requires controlled solder rework.

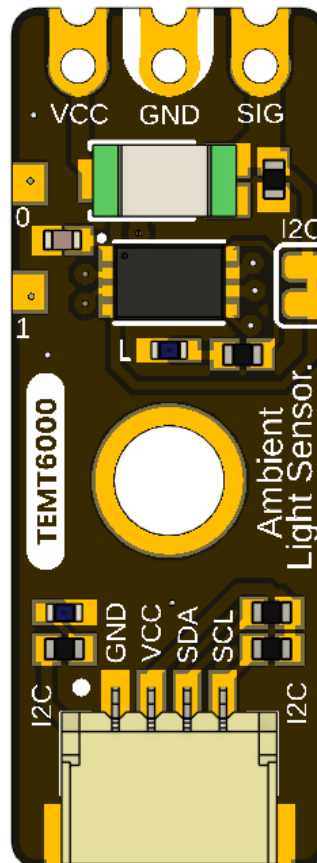
5.8 Troubleshooting Guide

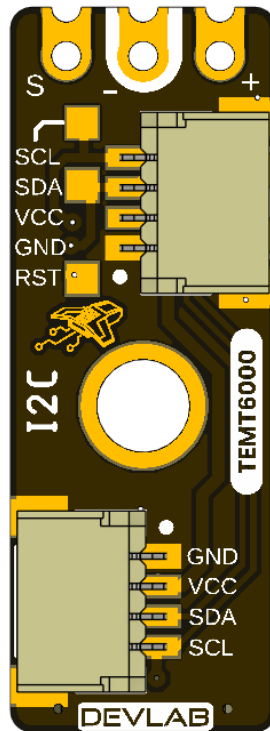
Symptom	Checks
No address found	Supply/bus-logic compatibility, ground, connector orientation, populated connector, SDA/SCL continuity, intact I2C bridge
Bus held low	Cable reversal, duplicate pull-ups, unintended drive on shared SDA/SWDIO or SCL/SWCLK, solder bridge, unpowered device
Address found but rejected	Verify DDP major 1, Device ID 0x0102, exact response lengths, and possible address collision
DDP identity succeeds but measurement fails	Check ANALOG_INPUT/SENSOR_DATA, 5 ms delay, exact length, byte order, and 12-bit range
Direct SIG always zero/full scale	Supply, ADC range, contact order, optical obstruction, current-board transfer function
Built-in LED unexpected	Use relay-compatible commands for PB5; digital-I/O commands are unsupported
Lux result inaccurate	Sensor spread, source spectrum, geometry, enclosure, temperature, ADC and calibration

6 Mechanical Information

The current package provides rendered top and bottom views for V0.3.1 but no controlled current-revision dimension drawing. Legacy V0.0.1 dimensions are archived under `hardware/resources/old/` and describe a materially different, shorter analog-only board.

6.1 Current Board Envelope





The board has a narrow module outline, direct contacts at one end, at least one central mounting feature, and optional horizontal Qwiic connector positions. Rendered views are suitable for identification, not production measurement.

6.2 Integration Clearances

Leave optical clearance around the transparent TEMT6000 sensor package and its field of view. Provide connector insertion and cable bend clearance at every populated Qwiic position. Keep conductive fasteners away from pads and service signals, and avoid enclosure features that shadow the sensor or LEDs.

6.3 Mechanical Data Not Specified

- Board length, width, thickness, tolerances, and finished mass
- Hole diameters, coordinates, and edge clearances
- Optional connector population variants and mating-part numbers
- Connector insertion envelope, retention force, and cable bend radius
- Maximum top- and bottom-side component heights
- Optical axis, keep-out cone, and enclosure-window specification
- PA0/PA1 and service-pad coordinates

Do not scale production dimensions from the images or reuse the archived V0.0.1 mechanical values for V0.3.1.

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

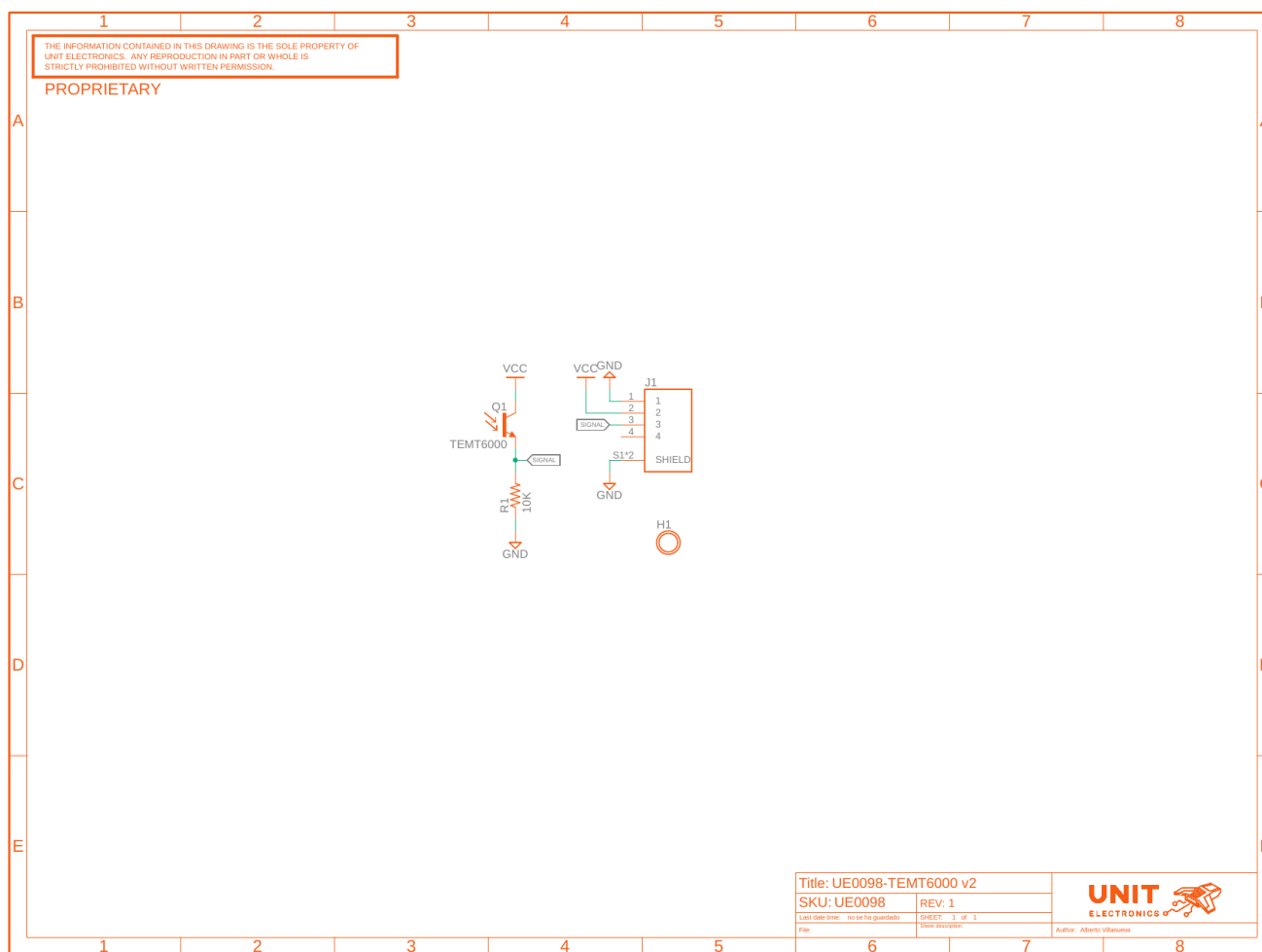
Reference	Location
Product repository	UNIT-Electronics-MX/unit_devlab_temt600_ambient_light_sensor
Hardware reference	hardware/README.md
English pinout	Released pinout — English
Spanish pinout	Released pinout — Spanish
Board top view	V0.3.1 resource
Board bottom view	V0.3.1 resource
Legacy schematic	Analog V0.0.1
Software examples	C++ and MicroPython
Local DDP device profile	TEMT6000 DDP profile
Canonical DDP definitions and Arduino library	UNIT-Electronics-Labs/unit_devlab_ddp_library
Comparative component page	Vishay TEMT6000X01 — reference only
Comparative component datasheet	Vishay Semiconductors, <i>TEMT6000X01 Ambient Light Sensor</i>, document 81579 — reference only

The released pinout and current board images define the visible interfaces. The legacy schematic defines only the earlier analog module. The Vishay datasheet is retained only as a comparative TEMT6000X01 profile; it does not establish the manufacturer or orderable part fitted to this module and does not define complete-module ratings. The DDP profile defines runtime identity and digital data transport; it does not replace the missing V0.3.1 electrical specification.

9 Appendix

9.1 Legacy V0.0.1 Schematic

The image below is rendered from hardware/unit_sch_V_0_0_1_ue0098_TEMT6000.pdf. It documents the former analog-only module and is retained for traceability. It is **not** the electrical schematic for the controller-based V0.3.1 board.



The legacy circuit connects a TEMT6000 phototransistor and 10 kΩ load to one analog signal. It contains no interface controller, SDA, SCL, PA0, PA1, reset, SWD, indicators, or I2C disable bridge.

9.2 Document Control

Field	Value
Product	UNIT ATOM TEMT6000 Ambient Light Sensor
Product hierarchy	UNIT Electronics → DevLab board ecosystem → Atom family
Manufacturer Part Number (MPN)	UE0098
Current board artwork	V0.3.1

Field	Value
Current pinout	V3.1.0
Available schematic	Legacy V0.0.1 only
Product Reference	Version 1.1.0
Publication date	2026-08-31
Interfaces	Qwiic I2C with shared factory SWD, analog, and auxiliary pads

9.3 Required Technical Releases

- V0.3.1 schematic and bill of materials
- Exact controller package/ordering suffix, released firmware image, and update procedure
- Exact oscillator/timing tolerances and uniform future DDP status behavior
- Complete-module supply/logic absolute-maximum ratings and current consumption
- Pull-up values and bus-loading limits
- Direct SIG transfer, loading, range, and ADC guidance
- Current-revision controlled mechanical drawing
- Module-level optical, electrical, environmental, and EMC characterization

9.4 Source Notes and Inconsistencies

- Current board images and pinout resources use different revision-number formats in their filenames and storage directories.
- Current pinout artwork calls the product DevLab: I2C TEMT6000; this reference identifies UNIT Electronics as the company that creates and develops the DevLab board ecosystem, with Atom as a product family within it.
- The current pinout shows an onboard controller and I2C/debug resources, but the only schematic in the repository is the earlier analog V0.0.1 circuit.
- Archived V0.0.1 dimensions, topology, and board views do not describe the longer V0.3.1 controller-based board.
- Qwiic positions A and B are marked optional; released assembly variants and controlled contact numbering are not supplied.
- Earlier examples treated the sensor as a digital threshold input. The sensor signal is analog; I2C operation is provided by the new onboard controller.
- Current firmware maps PA2 to ADC0 and PB5 to the relay-compatible actuator block. Digital-I/O commands 0x40..0x43 are not implemented; exposed PA0/PA1 remain unassigned.

9.5 Current Firmware Limitations

- The WATCHDOG capability is announced, but the current watchdog command does not operate an active hardware IWDG and must remain experimental.
- GET_RESET_INFO is not implemented and NRST behavior is incomplete.
- An internal ADC error can be returned as 0x0FFF, indistinguishable from a legitimate full-scale sample.
- Current main-loop code does not call the available prolonged-BUSY recovery.
- There is no sample timestamp, sample counter, or calibrated lux result.